Risk Analysis Home Assignment 2 Group 15

Problems 1 and 3 | AMI22Z

Al Amin Alif, Manthika Nethmini, and Nirmani Abisheka

Summary

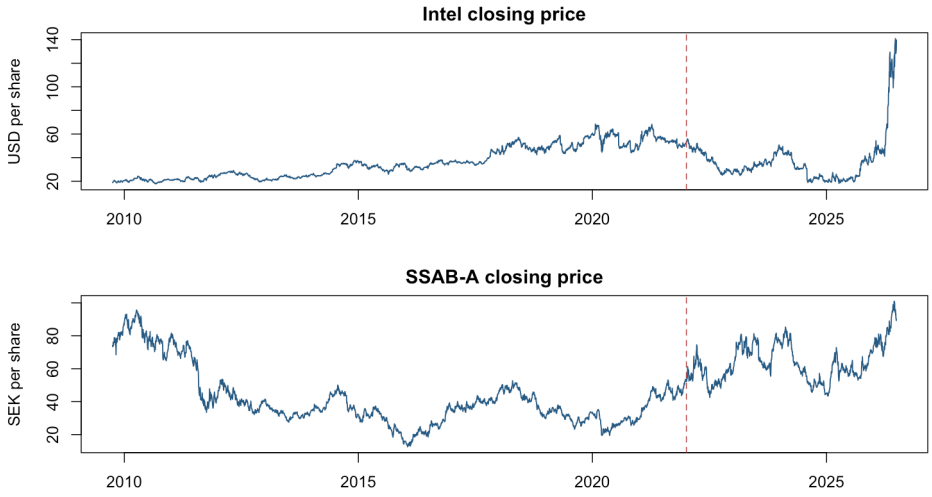
We assessed a $1,000 Intel and SSAB-A investment using 20,000 simulated daily price paths. The fitted closing-price model gives an expected profit of $481.87 and a 66.1% chance of profit; the observed price-only value on 30 June 2026 was $2,166.57 at the fixed purchase exchange rate. For the hospital, 40 beds redirect an estimated 12.36% of arriving patients. Simulations across capacities 40–70 identify 64 as the smallest capacity with daily occupancy at most 85% on at least 95% of days. These estimates depend on the stated models; simulation confidence intervals do not include uncertainty in their assumptions.

1 Problem 1 Stock portfolio forecast

1.1 Task 1 Historical closing prices

We used Yahoo Finance (2026) daily Close for Intel (INTC, USD) and SSAB-A (SSAB-A.ST, SEK), downloaded on 21 October 2026. The frozen CSVs contain both Close and Adj Close. Price observations begin on 1 October 2009. We fit the forecast using prices through 2021, then compare it with the closes on 30 June 2026. Figure 1 shows the two closing-price series. The dashed line marks the first trading day of 2022, 3 January, when the gift was invested. We chose 1 October 2009 as the starting date because it appears in the instructor's supplied StockMarket.R script, then applied it to the assigned assets.

Figure 1. Historical closing prices in each share's trading currency; dashed line is the investment date



1.2 Task 2 Daily log returns and parameter estimates

For each asset, a return dated t uses consecutive available closes. The first possible return is therefore 2 October 2009. We estimate the natural-log return mean by the sample average and variance by the sample variance with denominator n − 1:

$$r_t = \ln(P_t / P_{t-1}), \quad \hat{\mu} = (1/n)\Sigma r_t, \quad \hat{\sigma}^2 = \Sigma(r_t - \hat{\mu})^2/(n-1).$$

Table 1 gives the fitted parameters. The 95% intervals for μ use the t distribution; intervals for σ use the chi-square distribution (Olofsson & Andersson, 2012). These model-based intervals rely on approximately independent, identically distributed normal returns. They are separate from the simulation intervals for portfolio outcomes.

Table 1. Daily pre-2022 return parameters from closing prices

Asset	n returns	μ̂	σ²	σ̂	95% CI for μ	95% CI for σ
INTC	3,084	0.000325	0.000327	0.01808	[-0.000313, 0.000963]	[0.01764, 0.01854]
SSAB-A.ST	3,078	-0.000112	0.000604	0.02458	[-0.000981, 0.000756]	[0.02398, 0.02521]

Observed excess kurtosis is 15.5 for Intel and 3.9 for SSAB-A, so the normal-model interval for σ can understate uncertainty. As a diagnostic, 10,000 circular 20-day block resamples give 95% percentile intervals of [0.01590, 0.02077] and [0.02268, 0.02653] respectively. These intervals allow short-run dependence but still assume an approximately stable pre-2022 return process.

Both mean-return intervals include zero. The fitted long-run trend is therefore uncertain: over 1,134 days, the endpoints of the Intel mean interval imply median price factors of 0.70 to 2.98 under the model; for SSAB-A the corresponding factors are 0.33 to 2.36.

1.3 Task 3 Normal model and price trajectories

We model each daily return as normal with constant μ and σ^2. The 3,000 dates shared by the two return series give a same-day correlation of 0.273. We retain this dependence because the shares are held in one portfolio and common market shocks may affect both. US and Swedish closes are not simultaneous, so same-date correlation is only an approximation. Following the 252-day trading year in the brief (Dalarna University, 2026), the fitted bivariate normal model generates 20,000 complete paired price trajectories, each with 4.5 × 252 = 1,134 steps:

$$R_t = (r_{1,t}, r_{2,t})^T \sim N_2(m, V), \quad P_{j,t} = P_{j,0} \exp(\sum_{k=1}^{t} r_{j,k}), \; j = 1, 2.$$

Here j = 1 denotes Intel and j = 2 denotes SSAB-A. The vector m contains their fitted daily means; V has diagonal entries σ_1^2 and σ_2^2 and off-diagonal entries $\rho\sigma_1\sigma_2$.

Each path starts at the 3 January 2022 closes, $53.21 for Intel and SEK 52.58 for SSAB-A. At SEK 9.0447 per USD, $500 buys 9.3967 Intel shares, while $500 becomes SEK 4,522.37 and buys 86.0093 SSAB-A shares. Fractional shares are permitted. The forecast holds this purchase foreign-exchange (FX) rate fixed to isolate the specified share-price processes.

Figure 2. First 15 simulated share paths, model 5th–95th percentile band and median, and observed closes

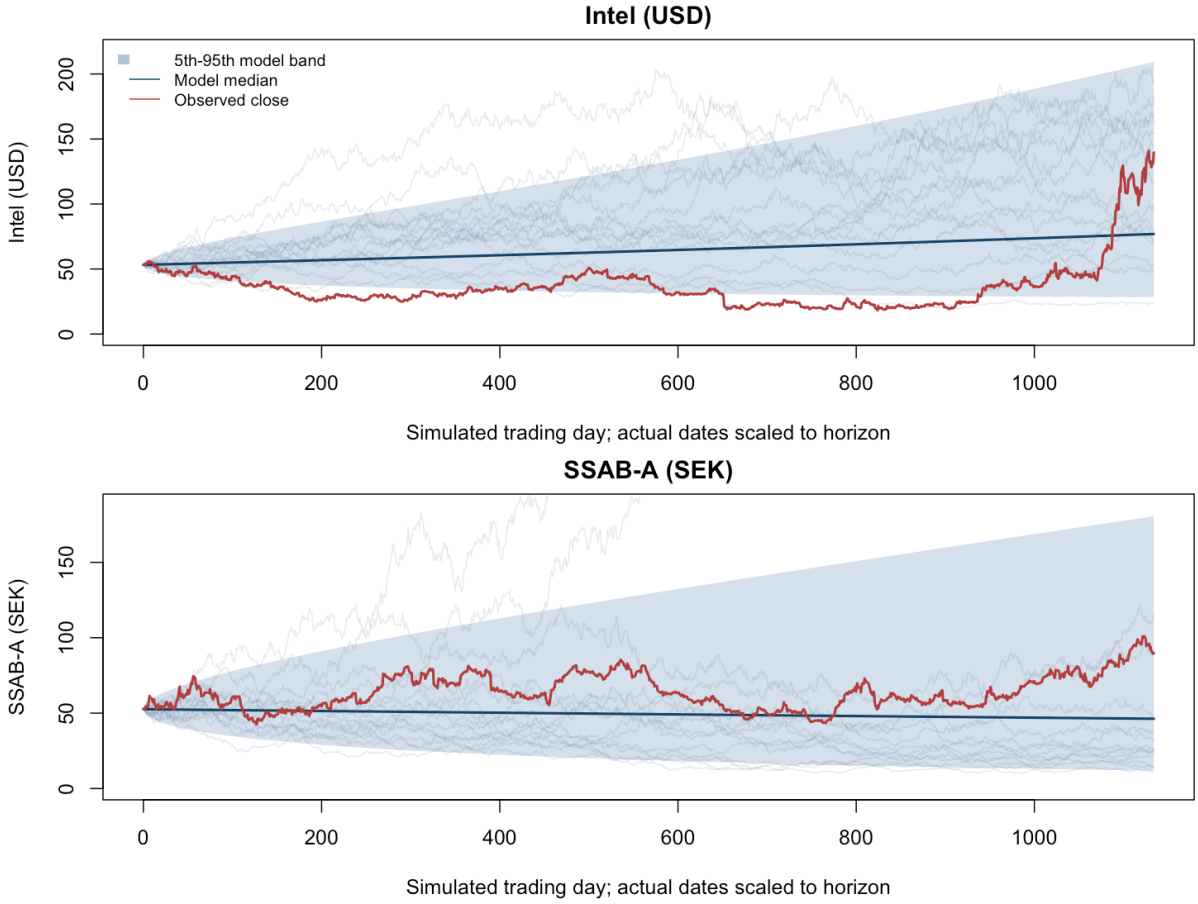


Figure 2 displays the two share-price trajectories in their own currencies. The shaded bands are pointwise model percentiles, not confidence intervals for the estimated mean. The vertical scale emphasizes the central band, so a few illustrated simulation paths extend beyond it. The observed paths have 1,125 actual return steps for each asset; their calendar dates are scaled across the 1,134-step forecast axis for display.

1.4 Task 4 Profitability and observed outcome

Answer: Yes, under the fitted price-only model the investment is profitable in expectation. The expected profit is $481.87 (95% Monte Carlo confidence interval $469.08–$494.65), and the simulated probability of finishing above $1,000 is 66.1%. A loss remains possible. Table 2 distinguishes the simulation mean, outcome range, and observed result.

Table 2. Forecast and observed portfolio values in USD

Measure	Result
Simulated mean final value	$1,481.87
Mean profit; 95% Monte Carlo CI	$481.87 [$469.08, $494.65]
Chance of profit; 95% Monte Carlo CI	66.1% [65.4%, 66.7%]
Median final value; 5th–95th outcome range	$1,259.18; $500.01–$3,193.00
Observed value, Close, fixed purchase FX	$2,166.57
Observed value, Close, actual USD cash-out	$2,108.11
Observed value, Adj Close proxy, fixed FX	$2,626.73
Observed value, Adj Close proxy, USD cash-out	$2,543.81

The observed price-only value at fixed FX is $2,166.57, a $1,166.57 profit and approximately the 83rd percentile of the simulated outcomes. Converting the SSAB holding back at the 30 June 2026 rate of SEK 9.7090 per USD gives a dollar cash-out of $2,108.11. The fixed-FX observed value is the like-for-like comparison with our fixed-FX forecast; the dollar cash-out also reflects exchange-rate movement.

Yahoo Adj Close is a total-return proxy that adjusts for distributions and corporate actions. Its purchase-to-final ratios imply an observed fixed-FX value of $2,626.73, versus $2,166.57 using Close. This proxy assumes reinvestment and is not a record of actual cash dividends received. We show it because distributions materially affect the realised economic return, while retaining the price-only comparison required by the closing-price model.

Figure 3. Distribution of simulated final values; the red line is the observed price-only value at fixed FX

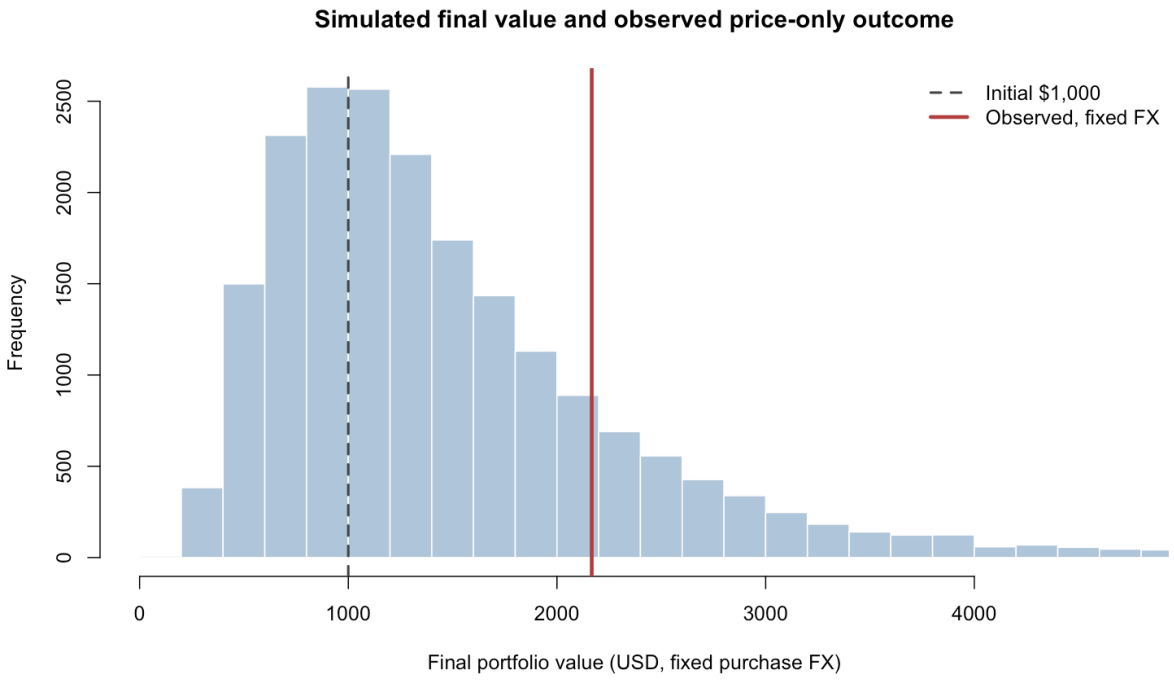


Figure 3 ends its horizontal axis at the simulated 99th percentile, leaving the highest 1% of outcomes outside the plotted range. As an analytic check on the Monte Carlo mean, the lognormal moment formula (Olofsson & Andersson, 2012) gives Σ_j $500 $\exp[1{,}134(\hat{\mu}_j + \sigma_j^2/2)]$ = $1,489.88, inside the simulation mean confidence interval.

The result depends strongly on the estimation period. Table 3 uses alternative pre-2022 windows and 100,000 terminal-value draws per scenario. The mean is calculated analytically for each fitted model. Every window gives positive expected profit, but the amount and profit probability differ substantially. Setting cross-share correlation to zero changes the portfolio distribution and raises the simulated profit probability from 66.0% to 69.4%; the analytic mean is unchanged.

Table 3. Sensitivity to pre-2022 estimation window and share-return correlation

Window	μ INTC	μ SSAB-A	ρ	Analytic mean final value	Simulated P(profit), 95% CI
2009–2021	0.000325	-0.000112	0.273	$1,489.88	66.0% [65.7%, 66.2%]
2012–2021	0.000295	0.000060	0.243	$1,595.93	69.9% [69.6%, 70.2%]
2017–2021	0.000271	0.000322	0.272	$1,847.55	75.9% [75.6%, 76.2%]
2019–2021	0.000119	0.000724	0.327	$2,383.49	81.8% [81.6%, 82.0%]
2009–2021, ρ = 0	0.000325	-0.000112	0.000	$1,489.88	69.4% [69.1%, 69.7%]

The baseline row repeats the main model with independent terminal draws. Its 66.0% and Table 2's 66.1% differ by Monte Carlo variation; their 95% intervals overlap.

These alternative windows are post hoc sensitivity checks, not precommitted forecasts. The 2009 start is tied to the instructor's starter script and provides the longest sample in our frozen dataset. Even so, the constant-parameter normal model omits changing volatility, heavy tails, possible regime shifts, transaction costs, taxes, inflation, and future FX variation. The Monte Carlo intervals measure simulation precision conditional on the fitted model, while the 5th–95th range describes potential outcomes under that model; neither includes uncertainty in its assumptions.

2 Problem 3 Hospital beds

2.1 Task 1 Daily occupancy at 40 beds

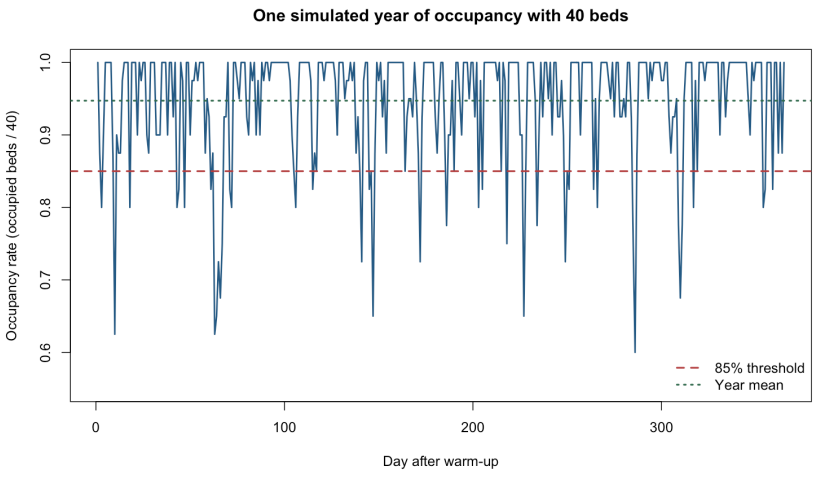
Let C be bed capacity and O_t the occupied beds after day t admissions. Patients stay at least one day. A geometric length of stay on {1, 2, ...} with mean 2.9 gives departure probability p = 1/2.9 between days. Each day we first retain the previous day's patients who have not left, then admit new arrivals up to capacity, then record occupancy. Departures free beds for the next day's intake, as stated in the brief (Dalarna University, 2026). This convention measures the occupied beds during the new intake day:

$$S_t \mid O_{t-1} \sim \text{Binomial}(O_{t-1}, 1 - 1/2.9), \quad A_t \sim \text{Poisson}(14.8)$$

$$O_t = \min(C, S_t + A_t), \quad L_t = \max(0, S_t + A_t - C).$$

Here L_t is the number redirected to another hospital; there is no queue. Recording occupancy after departures but before admitting arrivals would answer a different, less relevant question about temporary free beds. Figure 4 shows one 365-day path for C = 40 after a 365-day warm-up, so the initial empty hospital does not dominate the displayed year.

Figure 4. One year of daily occupied-bed fraction after warm-up at 40 beds; year mean 94.7% (long-run mean 94.0%)



2.2 Task 2 Probability of redirecting a patient

We ran 10,000 independent paths at each capacity tested, discarded 365 warm-up days per path, and measured the next 365 days. That supplies 3.65 million simulated days of the whole hospital per capacity. The patient rejection probability is total redirected arrivals divided by total arrivals, not the fraction of days with at least one rejection. Intervals use variation across independent paths, preserving dependence among days within a path.

Answer: With 40 beds, the estimated probability that a patient is redirected is 12.362% (95% simulation CI 12.340%–12.384%). An exact stationary calculation gives 12.370%, consistent with the simulation. Average occupied beds are 37.61. As a balance check, 14.8 × (1 − rejection probability) × 2.9 = 37.61 beds. Before rejections, the offered load is 14.8 × 2.9 = 42.92 beds, greater than the present capacity.

2.3 Task 3 Capacity meeting the occupancy condition

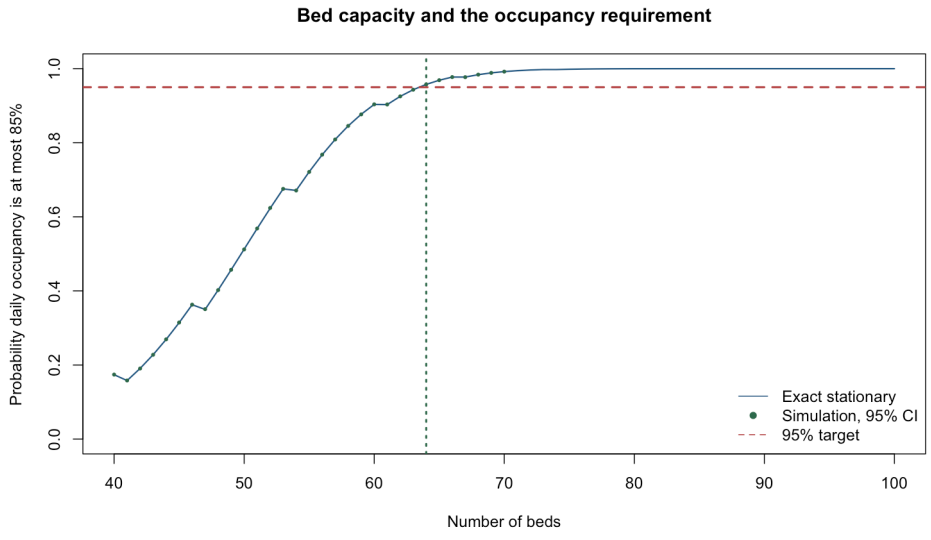
We simulated every integer capacity from 40 through 70, then selected the smallest simulated value with $Pr(O_t/C \leq 0.85) \geq 0.95$. We separately solved the finite-state stationary model for capacities 1 through 100 (Olofsson & Andersson, 2012), which confirms that no capacity below 40 qualifies and checks the simulated threshold. "Optimal" means the smallest capacity satisfying the stated probability condition, because the brief supplies no costs with which to optimise further.

Answer: 64 beds are required. At 63 beds the estimated qualifying-day probability is 94.32%; at 64 it is 95.76%. Their 95% simulation intervals lie on opposite sides of 95%. Table 4 provides the nearby capacities; Figure 5 shows the simulation against the stationary calculation.

Table 4. Long-run capacity results from 10,000 independent paths per capacity

Beds	Pr(occ ≤ 85%), 95% CI	Exact stationary probability	Simulated patient rejection
40	17.42% [17.36%, 17.47%]	17.38%	12.3619%
58	84.55% [84.48%, 84.61%]	84.54%	0.1863%
59	87.66% [87.60%, 87.71%]	87.69%	0.1303%
60	90.37% [90.32%, 90.42%]	90.35%	0.0870%
61	90.34% [90.29%, 90.39%]	90.30%	0.0587%
62	92.52% [92.48%, 92.56%]	92.52%	0.0384%
63	94.32% [94.28%, 94.36%]	94.32%	0.0255%
64	95.76% [95.72%, 95.79%]	95.75%	0.0157%
65	96.89% [96.86%, 96.91%]	96.87%	0.0103%
66	97.73% [97.71%, 97.75%]	97.73%	0.0065%

Figure 5. Capacity search: simulation points with 95% intervals (narrower than the points), exact stationary curve, and 95% target



The exact and simulated probabilities differ by at most 0.092 percentage points across the 40–70 grid. The curve can dip when one bed is added: the allowed occupied count ⌊0.85C⌋ sometimes stays unchanged, while the extra bed reduces blocking and shifts occupancy upward. For this reason we tested integer capacities rather than interpolating the curve. The result assumes constant Poisson demand, independent arrivals, identical geometric stays, and no waiting room; weekday patterns, seasonality, emergency priorities, and bed costs could change a real planning decision.

3 Reproducibility

The companion file Group_15_Analysis_and_Data.zip contains the base-R script analysis.R, the three frozen input CSVs in raw/, and the generated results. The script fixes random seeds: 150103 for the main stock simulation, 160101 for window sensitivity, 3141 and 3142 for the block bootstrap, 1000 + C for each capacity, and 1540 for the displayed occupancy path. The main forecast simulates full daily trajectories; the auxiliary window comparison uses the equivalent terminal distribution of summed normal returns. The reported tables and figures come from our 31 October 2026 run in R 4.5.2; the numerical outputs also reproduce under R 4.3.3. The analysis uses R (R Core Team, 2025).

AI assistance declaration

After our initial attempt to complete the entire assignment without AI, we sought OpenAI's help to refine our approach, organise the source data, test and improve the analysis code, evaluate the figures, and polish this report. We ran the script in R 4.5.2 to generate the reported results. The group is responsible for reviewing and submitting the final version.

References and data

Dalarna University. (2026). Risk Analysis (AMI22Z), Home Assignment 2 [assignment brief].

Olofsson, P., & Andersson, M. (2012). Probability, Statistics, and Stochastic Processes (2nd ed.). Wiley.

R Core Team. (2025). R: A language and environment for statistical computing. R Foundation for Statistical Computing. https://www.R-project.org/

Yahoo Finance. (2026). Historical daily Close and Adj Close for INTC and SSAB-A.ST, and USD/SEK (SEK=X), retrieved 21 October 2026. https://finance.yahoo.com/quote/INTC/history/ ; https://finance.yahoo.com/quote/SSAB-A.ST/history/ ; https://finance.yahoo.com/quote/SEK=X/history/ . Frozen CSV and JSON files accompany the R script.